

worked problems
wed 9 + TH 9

2.4. Applications of Linear First-Order Differential Equations

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In Exercises 14–20, solve each of the given nonhomogeneous linear first-order DEs.

14. $y' + y = 2$

15. $y' + 2y = 2t$

16. $y' + ty = 10t$

17. $y' + \frac{2}{t}y = e^t$

18. $y' = -(y - 1) \cot t$

19. $(1 + t^2)y' + 2ty = 2t$

20. $y' = 0.03 - \frac{2}{100 - t}y$

In Exercises 21–27, solve each of the given initial-value problems.

21. $y' + y = 2, \quad y(0) = 3$

22. $y' + 2y = 2t, \quad y(1) = 0$

23. $y' + ty = 10t, \quad y(0) = 5$

24. $y' + \frac{2}{t}y = e^t, \quad y(1) = 4, \quad t > 0$

✓ 25. $y' = -(y - 1) \cot t, \quad y(\pi/2) = 1, \quad 0 < t < \pi$

sep

26. $(1 + t^2)y' + 2ty = 2t, \quad y(0) = 1$

27. $y' = 0.03 - \frac{2}{100 - t}y, \quad y(0) = 1$

2.4 Applications of Linear First-Order Differential Equations

A large number of important physical situations can be modeled by linear first-order differential equations. In this section we introduce several such applications through examples and explore further scenarios in the exercises.

1. $y' = 10y$

2. $y' = 10y + 10$

3. $y' = 10y^2$

4. $y' = 10y^2 - 10$

✓ 5. $t^2y' + y^2 = 1$

6. $e^{3t+y} \frac{dy}{dt} = 1$

7. $t dy - (y - 1) dt = 0$

8. $\frac{dy}{dt} = \frac{5ty - t}{4 + t^2}$

✓ 9. $y - t \frac{dy}{dt} = 6 - 3t^2 \frac{dy}{dt}$

✓ 10. $\frac{dy}{dt} = \frac{-2ty}{t^2 + 1}$

11. $(2 + t^2)y' + 2ty = 0$

✓ 12. $3y^2y' + t^2 = 0$

✓ 13. $(y + t)y' + y = t$

14. $y' \sin 2t + 2y \cos 2t = 0$

Solve each of the DEs in Exercises 15–28.

15. $y' = 10y$

16. $y' = 10y + 10$

17. $y' = 10y^2$

18. $y' = 10y^2 - 10$

19. $t^2y' + y^2 = 1$

✓ 20. $e^{3t+y} \frac{dy}{dt} = 1$

21. $t dy - (y - 1) dt = 0$

22. $\frac{dy}{dt} = \frac{5ty - t}{4 + t^2}$

✓ 23. $y - t \frac{dy}{dt} = 6 - 3t^2 \frac{dy}{dt}$

24. $\frac{dy}{dt} = \frac{-2ty}{t^2 + 1}$

25. $(2 + t^2)y' + 2ty = 0$

26. $3y^2y' + t^2 = 0$

27. $(y + t)y' + y = t$

28. $y' \sin 2t + 2y \cos 2t = 0$

Solve each of the IVPs stated in Exercises 29–42.

29. $y' = 10y, \quad y(0) = 3$

✓ 30. $y' = 10y + 10, \quad y(0) = 2$

31. $y' = 10y^2, \quad y(1) = 4$

32. $y' = 10y^2 - 10, \quad y(1) = -1$

33. $t^2y' + y^2 = 1, \quad y(2) = 0$

34. $e^{3t+y} \frac{dy}{dt} = 1, \quad y(0) = 0$

35. $t dy - (y - 1) dt = 0, \quad y(1) = 3$

36. $\frac{dy}{dt} = \frac{5ty - t}{4 + t^2}, \quad y(1) = 1$

37. $y - t \frac{dy}{dt} = 6 - 3t^2 \frac{dy}{dt}, \quad y(1) = 5$

38. $\frac{dy}{dt} = \frac{-2ty}{t^2 + 1}, \quad y(0) = 4$

39. $(2 + t^2)y' + 2ty = 0, \quad y(1) = 1$

40. $3y^2y' + t^2 = 0, \quad y(0) = 1$

41. $(y + t)y' + y = t, \quad y(0) = 1$

wed

P121
25

$$\frac{dy}{dt} = -(y-1) \frac{\cos(t)}{\sin(t)} \quad \text{sep}$$

$$\int \frac{dy}{y-1} = \int \frac{-\cos(t)}{\sin(t)} dt + C$$

$$\ln|y-1| = - \int \frac{1}{u} du + C \quad \begin{matrix} u = \sin(t) \\ du = \cos(t) \end{matrix}$$

$$\ln|y-1| = -\ln|\sin(t)| + C$$

or



$$\frac{dy}{dt} + y \frac{\cos(t)}{\sin(t)} = \frac{\cos(t)}{\sin(t)}$$

$$v(t) = e^{\ln(\sin(t))} = \sin(t)$$

$$\sin(t) y' + \cos(t) y = \cos(t)$$

$$(y \sin(t))' = \cos(t)$$

$$y \sin(t) = \sin(t) + C$$

p137
20

$$e^{3t} \cdot e^y \frac{dy}{dt} = 1 \quad \text{sep}$$

$$\int e^y dy = \int e^{-3t} dt + C$$

$$e^y = \frac{e^{-3t}}{-3} + C$$

p138
23

~~At the~~

$$y + (3t^2 - t) \frac{dy}{dt} = 6$$

lin

$$\frac{dy}{dt} + \frac{1}{3t^2 - t} y = \frac{6}{3t^2 - t}$$

$$v(t) = e^{\int \frac{1}{3t^2 - t} dt} \quad \text{☹️}$$

sep

$$(3t^2 - t) \frac{dy}{dt} = 6 - y$$

$$\int \frac{dy}{6-y} = \int \frac{dt}{3t^2 - t} + C$$

$$-\ln(6-y) = \int \frac{dt}{3t^2 - t} + C$$

Int continued on next page

Int for #23 (euler way)

$$\int \frac{dt}{t(t-3)} = \int \frac{A}{t} dt + \int \frac{B}{t-3} dt$$

where $A/t + B/t-3 = 1/t(t-3)$

~~$$A(t-3) + Bt = 1$$~~

$$A(t-3) + Bt = 1$$

$$(A+B)t - 3A = 0t + 1$$

$$-3A = 1 \quad A = -1/3$$

$$A+B=0 \quad B = -A = 1/3$$

$$\begin{aligned} \int \frac{dt}{t(t-3)} &= -\frac{1}{3} \ln(t) + \frac{1}{3} \ln(t-3) \\ &= \frac{1}{3} (\ln(t-3) - \ln(t)) \end{aligned}$$

$$\int \frac{1}{t(t-3)} dt = \frac{1}{3} \ln(t-3/t) = \ln \left(\frac{t-3}{t} \right)^{1/3}$$

p137

TH

9

~~TH~~

$$y - t y' = 6 - 3t^2 y'$$

check exact $dt y - t dy = 6 dt - 3t^2 dy$

$$(y-6)dt + (3t^2-t)dy = 0$$

not exact $\frac{\partial}{\partial y}(y-6) = 1 \neq \frac{\partial}{\partial t}(3t^2-t)$

TH

p138
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$$y' - 10y = 10 \quad y(0) = 2 \quad \text{lin}$$

$$e^{-10t} y' - 10e^{-10t} y = 10e^{-10t} \quad v = e^{-10t}$$

$$y e^{-10t} = \int 10 e^{-10t} dt + C$$

$$u = -e^{-10t} + C$$

$$2e^0 = -e^0 + C \Rightarrow \boxed{C=3}$$

$$\boxed{y e^{-10t} = -e^{-10t} + 3}$$

p137

Th

10

$$dy/dt = -2ty/(t^2+1)$$

check exact

$$2ty dt + (t^2+1) dy = 0$$

$$\frac{\partial}{\partial y}(2ty) = 2t = \frac{\partial}{\partial t}(t^2+1) \quad \text{Exact.}$$

$$\frac{\partial s}{\partial t} = 2ty \Rightarrow s = t^2 y + h(y)$$

$$\Rightarrow \frac{\partial s}{\partial y} = t^2 + h'(y) = t^2 + 1$$

$$h'(y) = 1 \quad h(y) = y$$

$$\boxed{t^2 y + y = C}$$

■ Solve $y' = y + ty^2$

$$\boxed{\begin{aligned} u &= y^{-1} \\ u' &= -y^{-2} y' \end{aligned}}$$

$$\frac{e^t}{y} = C - \left[te^t - e^t \right]$$

$$y' - y = ty^2 \Rightarrow y^{-2} y' - y^{-2} y = t$$

$$y^{-2} y' - y^{-1} = t$$

$$-u' - u = t$$

$$e^t (u' + u) = -t$$

$$(e^t u)' = -te^t$$

$$e^t u = \int -te^t dt + C$$

2. Solve $y' + y = y^3$

$$y^{-3} y' + y y^{-3} = 1$$

$$y^{-3} y' + y^{-2} = 1$$

$$-\frac{1}{2} u' + u = 1$$

$$u' - 2u = -2 \quad \text{lin} \quad v = e^{-2t}$$

$$e^{-2t} u' - 2ue^{-2t} = -2e^{-2t}$$

$$(e^{-2t} u)' = -2e^{-2t}$$

$$e^{-2t} u = -2 \int e^{-2t} dt + C$$

$$e^{-2t} u = e^{-2t} + C$$

$$\boxed{e^{-2t} y^{-2} = e^{-2t} + C}$$

$$u = t \quad dv = e^t dt$$

$$\boxed{\int u dv = uv - \int v du}$$

$$du = dt \quad v = e^t$$

$$u = y^{-2}, \quad u' = -2y^{-3} y'$$

p137

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$$3y^2 y' + t^2 = 0$$

$$m dt + n dy = 0$$

$$3y^2 dy + t^2 dt = 0$$

$$t^2 dt + 3y^2 dy = 0$$

$$\frac{\partial}{\partial y}[t^2] = 0 = \frac{\partial}{\partial t}[3y^2]$$

Exact!

$$\frac{\partial s}{\partial t} = t^2 \Rightarrow s(t, y) = \frac{t^3}{3} + h(y)$$

 $\Downarrow$

$$\frac{\partial s}{\partial y} = 0 + h'(y) = 3y^2$$

 $\Downarrow$

$$h'(y) = 3y^2$$

 $\Downarrow$

$$h(y) = y^3$$

$$\boxed{\frac{t^3}{3} + y^3 = c}$$

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p137

$$t^2 y' + y^2 = 1 \quad \text{sep}$$

$$t^2 \frac{dy}{dt} = 1 - y^2$$

$$\int \frac{dy}{1-y^2} = \int t^{-2} dt + C$$

$$\int \frac{1}{(1-y)(1+y)} dy = -t^{-1} + C$$

$$\frac{1}{(1-y)(1+y)} = \frac{A}{1-y} + \frac{B}{1+y} \quad \text{partial Fractions}$$

$$0y + 1 = A(1+y) + B(1-y) = (A+B) + y(A-B)$$

$$A+B=1 \quad A=B=\frac{1}{2}$$

$$A-B=0$$

$$-\frac{1}{2} \ln(1-y) + \frac{1}{2} \ln(1+y) = -t^{-1} + C$$

$$\int \frac{\frac{1}{2}}{1-y} dy = -\frac{1}{2} \int \frac{1}{u} du = -\frac{1}{2} \ln(1-y)$$

$$u=1-y$$

$$du = -dy$$

p137

#13

$$(y+t)dy + ydt = tdt$$

$$(y-t)dt + (y+t)dy = 0$$

$$\frac{\partial}{\partial y}[y-t] = 1$$

$$1 = \frac{\partial}{\partial t}[y+t]$$

Exact

$$\frac{\partial s}{\partial t} = y-t$$

$$\frac{\partial s}{\partial y} = y+t$$



$$s(t, y) = \int (y-t) dt = yt - \frac{t^2}{2} + h(y)$$



$$\frac{\partial s}{\partial y} = t - 0 + h'(y) = y+t$$

$$h'(y) = y \Rightarrow h(y) = \frac{y^2}{2}$$

$$yt - \frac{t^2}{2} + \frac{y^2}{2} = C$$

POP

POP

immortal
cockroaches

50% female

~~150~~ 150 Babies/year
per female. ^(50:50)

$$r'(t) = \frac{150}{2} r(t)$$

r = female
cockroaches

$$r(t) = r(0) e^{75t}$$

mortal cockroaches
2 year life expectancy

$$r'(t) = \left(\frac{150}{2} - \frac{1}{2} \right) r(t) \quad \text{ODE}$$

$$r(t) = r(0) e^{74.5t} \quad \text{SOL}$$

$$\frac{dr}{dt} = 74.5 r \Rightarrow \left(\frac{dr}{r} = 74.5 dt + C \right)$$

not
needed

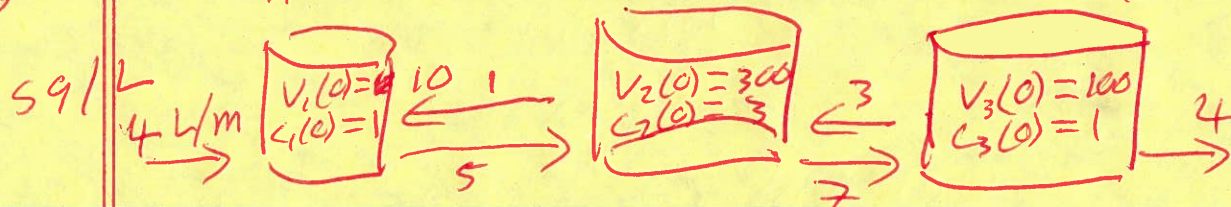
$$\Rightarrow \ln(r) = 74.5t + C$$

$$\Rightarrow r = e^{74.5t} \cdot e^C$$

$$r = r(0) e^{74.5t}$$

TANKS

$$S_1' = 20 - 5C_1 + 1C_2 + 0C_3 = 20 = 5/10 S_1 + 1/300 S_2 + 0 S_3$$



$$y' = Ay + S \quad y(0) = y_0$$

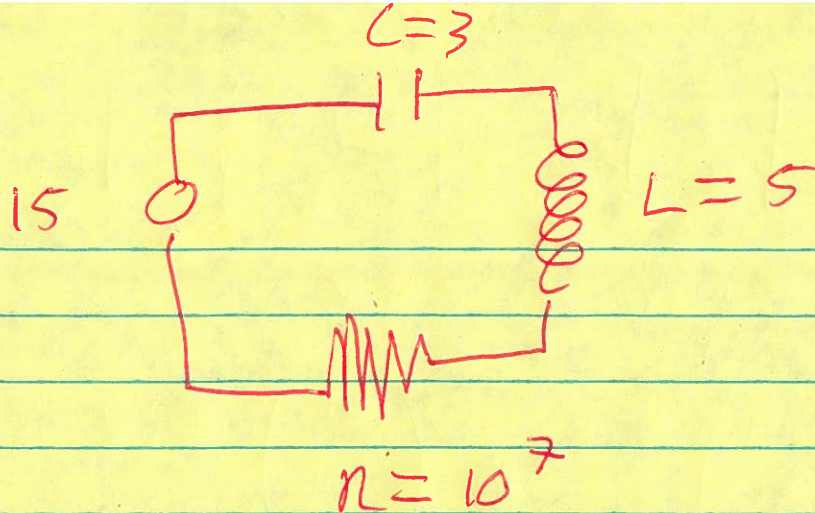
$$y_0 = \begin{pmatrix} 10 \\ 900 \\ 100 \end{pmatrix}$$

$$S = \begin{pmatrix} 20 \\ 0 \\ 0 \end{pmatrix}$$

$$A = \begin{pmatrix} -5/10 & 1/300 & 0 \\ +5/10 & -8/300 & +3/100 \\ 0 & +7/300 & -7/100 \end{pmatrix}$$

~~8/100~~

Circuits



write down ODE

$$15 + \frac{q}{3} + 5q'' + 10^7 q' = 0$$